

Mathematical Foundations of Political Methodology

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- Multivariate Optimization with Constraints

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Multivariate Optimization

Optimizing multivariate functions

- Parameters $\beta = (\beta_1, \beta_2, \dots, \beta_n)$ such that $f(\beta|\mathbf{X}, \mathbf{Y})$ is maximized
- Policy $\mathbf{x} \in \mathbb{R}^n$ that maximizes $U(\mathbf{x})$
- Weights $\pi = (\pi_1, \pi_2, \dots, \pi_K)$ such that a weighted average of forecasts $\mathbf{f} = (f_1, f_2, \dots, f_k)$ have minimum loss

$$\min_{\pi} = -\left(\sum_{j=1}^K \pi_j f_j - y\right)^2$$

Today we'll describe analytic and computational approaches to optimization

- Analytic recipe for optimization
- Computational optimization
 - Multivariate Newton-Raphson
 - BFGS
 - Approximate Optimization: k-means

Multivariate Optimization

Definition

Let $\mathbf{x} \in \mathbb{R}^n$ and let $\delta > 0$. Define a *neighborhood* of $\mathbf{x}$, $B(\mathbf{x}, \delta)$, as the set of points such that,

$$B(\mathbf{x}, \delta) = \{\mathbf{y} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{y}\| < \delta\}$$

Definition

Suppose $f : X \rightarrow \mathbb{R}$ with $X \subset \mathbb{R}^n$. A vector $\mathbf{x}^* \in X$ is a *global maximum* if, for all other $\mathbf{x} \in X$

$$f(\mathbf{x}^*) > f(\mathbf{x})$$

A vector $\mathbf{x}^{local}$ is a *local maximum* if there is a neighborhood around $\mathbf{x}^{local}$, $Q \subset X$ such that, for all $\mathbf{x} \in Q$,

$$f(\mathbf{x}^{local}) > f(\mathbf{x})$$

Multivariate Optimization

Definition

A set $X \subset \mathbb{R}^n$ is *compact* if it is closed and bounded

Theorem

Multivariate Extreme Value Theorem Suppose $f : X \rightarrow \mathbb{R}$ be continuous and $X \subset \mathbb{R}^n$ and X compact. Then f takes on its *maximum* and *minimum* values on X .

We're going to come up with the multivariate equivalent of the *first order* and *second order* conditions now

Gradient

Definition

Suppose $f : X \rightarrow \mathbb{R}^n$ with $X \subset \mathbb{R}^1$ is a differentiable function. Define the gradient vector of f at $\mathbf{x}_0$, $\nabla f(\mathbf{x}_0)$ as,

$$\nabla f(\mathbf{x}_0) = \left(\frac{\partial f(\mathbf{x}_0)}{\partial x_1}, \frac{\partial f(\mathbf{x}_0)}{\partial x_2}, \frac{\partial f(\mathbf{x}_0)}{\partial x_3}, \dots, \frac{\partial f(\mathbf{x}_0)}{\partial x_n} \right)$$

Gradient First Order Condition

Theorem

Suppose $f : X \rightarrow \mathbb{R}^1$, $X \subset \mathbb{R}^n$. Suppose $\mathbf{a} \in X$ is a *local* extremum. Then,

$$\begin{aligned}\nabla f(\mathbf{a}) &= \mathbf{0} \\ &= (0, 0, \dots, 0)\end{aligned}$$

- Proof (intuition): same as one dimensional case (left-hand, right hand), just do it dimension by dimension
- **Critical Values:**
 - 1) Maximum
 - 2) Minimum
 - 3) **Saddle point**
- **Second Derivative Test!**

Second Order Conditions: Hessian

Definition

Suppose $f : X \rightarrow \mathbb{R}^1$, $X \subset \mathbb{R}^n$, with f a twice differentiable function. We will define the **Hessian** matrix as the matrix of second derivatives at $\mathbf{x}^* \in X$,

$$H(f)(\mathbf{x}^*) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}^*) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f}{\partial x_2 \partial x_2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{x}^*) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n}(\mathbf{x}^*) \end{pmatrix}$$

General test $\rightsquigarrow$ Two Dimensional Test $\rightsquigarrow$ Example

Hessians

Definition

Consider $n \times n$ matrix $\mathbf{A}$. If, for all $\mathbf{x} \in \mathbb{R}^n$ where $\mathbf{x} \neq 0$:

$$\mathbf{x}' \mathbf{A} \mathbf{x} > 0 \text{ } \mathbf{A} \text{ is positive definite}$$

$$\mathbf{x}' \mathbf{A} \mathbf{x} < 0 \text{ } \mathbf{A} \text{ is negative definite}$$

If $\mathbf{x}' \mathbf{A} \mathbf{x} > 0$ for some $\mathbf{x}$ and $\mathbf{x}' \mathbf{A} \mathbf{x} < 0$ for other $\mathbf{x}$, then we say $\mathbf{A}$ is *indefinite*

Intuition for Quadratic Form

Suppose $\mathbf{x}^*$ is some critical value,

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) + (\mathbf{x} - \frac{1}{2}\mathbf{x}^*)\mathbf{H}(f)(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) + R(\mathbf{x}^*, \mathbf{x})$$

$$f(\mathbf{x}) - f(\mathbf{x}^*) = 0(\mathbf{x} - \mathbf{x}^*) + (\mathbf{x} - \frac{1}{2}\mathbf{x}^*)\mathbf{H}(f)(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) + R(\mathbf{x}^*, \mathbf{x})$$

For $\mathbf{x}$ near $\mathbf{x}^*$, $R(\mathbf{x}^*, \mathbf{x}) \approx 0$

$\mathbf{H}(f)(\mathbf{x}^*)$ positive definite $\rightarrow f(\mathbf{x}) > f(\mathbf{x}^*) \rightarrow$ local minimum

$\mathbf{H}(f)(\mathbf{x}^*)$ negative definite $\rightarrow f(\mathbf{x}) < f(\mathbf{x}^*) \rightarrow$ local maximum

Theorem

Second Derivative Test

- If $\mathbf{H}(f)(\mathbf{a})$ is *positive definite* then $\mathbf{a}$ is a local minimum
- If $\mathbf{H}(f)(\mathbf{a})$ is *negative definite* then $\mathbf{a}$ is a local maximum
- If $\mathbf{H}(f)(\mathbf{a})$ is *indefinite* then $\mathbf{a}$ is a saddle point

Second Derivative Test

Many ways to assess definiteness $\rightsquigarrow$ use determinant

Theorem

Two Dimensional, Second Derivative Test. Suppose $f : X \rightarrow \mathbb{R}$ with $X \subset \mathbb{R}^2$ and f twice differentiable. Write the *Hessian* of f at a critical value $\mathbf{a}$,

$$\mathbf{H}(f)(\mathbf{a}) = \begin{pmatrix} A & B \\ B & C \end{pmatrix}$$

Then, we can conduct the second derivative test as:

- $AC - B^2 > 0$ and $A > 0 \rightsquigarrow$ *positive definite* $\rightsquigarrow \mathbf{a}$ is a local minimum
- $AC - B^2 > 0$ and $A < 0 \rightsquigarrow$ *negative definite* $\rightsquigarrow \mathbf{a}$ is a local maximum
- $AC - B^2 < 0 \rightsquigarrow$ *indefinite* $\rightsquigarrow$ saddle point
- $AC - B^2 = 0$ *inconclusive*

Multivariate Recipe

- 1) Calculate **gradient**
- 2) Set equal to zero, solve system of equations
- 3) Calculate **Hessian**
- 4) Assess **Hessian** at critical values
- 5) **Boundary values?** (if relevant)

Example 1: A Simple Optimization Problem

Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ with

$$f(x_1, x_2) = 3(x_1 + 2)^2 + 4(x_2 + 4)^2$$

Calculate gradient

$$\nabla f(\mathbf{x}) = (6x_1 + 12, 8x_2 + 32)$$

$$\mathbf{0} = (6x_1^* + 12, 8x_2^* + 32)$$

We now solve the system of equations to yield $x_1^* = -2$ and $x_2^* = -4$

Example 1: A Simple Optimization Problem

$$\mathbf{H}(f)(\mathbf{x}^*) = \begin{pmatrix} 6 & 0 \\ 0 & 8 \end{pmatrix}$$

$\det(\mathbf{H}(f)(\mathbf{x}^*)) = 48$ and $6 > 0$ so $\mathbf{H}(f)(\mathbf{x}^*)$ is positive definite. **local minimum**

Example 2: Two Dimensional Ideal Points

Suppose legislators are considering legislation $\mathbf{x} \in \mathbb{R}^2$. And suppose legislator i has utility function $U_i : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$U(\mathbf{x})_i = -(x_1 - \mu_1)^2 - (x_2 - \mu_2)^2$$

What is legislator i 's **optimal** policy?

$$\nabla f(\mathbf{x}) = (-2(x_1 - \mu_1), -2(x_2 - \mu_2))$$

$$\nabla f(\mathbf{x}) = \mathbf{0}$$

$$-2(x_1^* - \mu_1) = 0$$

$$-2(x_2^* - \mu_2) = 0$$

Solving yields $x_1^* = \mu_1$ and $x_2^* = \mu_2$.

Example 2: Two Dimensional Ideal Points

$$U(\mathbf{x})_i = -(x_1 - \mu_1)^2 - (x_2 - \mu_2)^2$$

Call $\boldsymbol{\mu} = (\mu_1, \mu_2)$

The Hessian at the critical value is

$$\begin{aligned} \mathbf{H}(f)(\boldsymbol{\mu}) &= \begin{pmatrix} \frac{\partial^2 U_i}{\partial x_1 \partial x_1}(\boldsymbol{\mu}) & \frac{\partial^2 U_i}{\partial x_1 \partial x_2}(\boldsymbol{\mu}) \\ \frac{\partial^2 U_i}{\partial x_2 \partial x_1}(\boldsymbol{\mu}) & \frac{\partial^2 U_i}{\partial x_2 \partial x_2}(\boldsymbol{\mu}) \end{pmatrix} \\ &= \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix} \end{aligned}$$

So, $-2 * -2 - 0 = 4 > 0$ and $-2 < 0 \rightsquigarrow$ **negative definite**, maximum $\boldsymbol{\mu} = (\mu_1, \mu_2)$ are legislator i 's two dimensional ideal point.

- Multivariate Optimization
- Multivariate Optimization with Constraints

Optimizing with constraints

We wish to maximize the function $f(x, y)$ subject to the constraint that $g(x, y) = 0$

With two unknowns (x, y) we could solve for $g(x, y) = 0$ in terms of y , substitute back into $f(x, y)$, and then optimize

Example: Maximize $U(x, y) = x^{\frac{1}{3}}y^{\frac{2}{3}}$ subject to $W = 5x + 4y$

This could be reexpressed as “Maximize $U(x, \frac{W-5x}{4})$ ”

This type of substitution isn't always feasible, particularly with more than two variables

Optimizing with constraints

We want to maximize $f(x, y)$ subject to $g(x, y) = 0$

We start by defining the *Lagrangian* L as a function of three variables: x , y , and λ (the Lagrange multiplier):

$$L(x, y, \lambda) = f(x, y) - \lambda g(x, y)$$

(Given a few assumptions on f and g) any solution to the constrained optimization problem is a critical point of the Lagrangian, and so solves

$$\frac{\partial L}{\partial x} = 0 \quad \frac{\partial L}{\partial y} = 0 \quad \frac{\partial L}{\partial \lambda} = 0$$

Optimizing with constraints

The steps

- 1 Introduce λ and form the Lagrangian L
- 2 Look at the critical points of L : a critical point (in x and y) will be a constrained maximum of the original problem
- 3 If there are multiple critical points, examine them by hand to figure out which type of points they are (using a *bordered Hessian*—a topic we haven't covered—or a concavity argument, or some other ad hoc method like a diagram)

Optimizing with constraints

Example: Maximize $f(x, y) = xy$ subject to $5x + 2y = 10$

$$L(x, y, \lambda) = xy - \lambda(5x + 2y - 10)$$

$$\nabla L = \begin{bmatrix} y - 5\lambda \\ x - 2\lambda \\ -5x - 2y + 10 \end{bmatrix} = \mathbf{0}$$

Solving for x, y, λ : $y = 5\lambda$, $x = 2\lambda$,

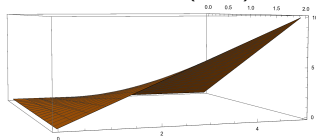
$$10\lambda + 10\lambda - 10 = 0$$

and so $\lambda = \frac{1}{2}$ and $x = 1$ and $y = \frac{5}{2}$

Optimizing with constraints

We can plot to confirm $x = 1, y = \frac{2}{5}$ is a constrained (global) max

The function $f(x, y) = xy$



The function $f(x, y) = xy$ and the constraint $y = \frac{10-5x}{2}$

